# Momen Hamza

AI & Machine Learning Engineer | Computer Vision · NLP · Data Science & Analytics

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## SUMMARY

Artificial Intelligence and Data Science graduate specializing in deep learning, computer vision, and NLP, with growing depth in applied AI engineering and data analytics. Hands-on experience designing and deploying end-to-end ML systems — from CNN- and transformer-based models to real-time, multimodal, and multilingual applications — and building production-style pipelines (RAG, vector databases, knowledge graphs, FastAPI + Docker). Comfortable across the full data workflow — from SQL analytics, statistics, EDA, and BI dashboards through to model training and live deployment. Passionate about building practical AI that solves real-world problems.

## EDUCATION

**Tafila Technical University** Sep 2020 – Jan 2025

*B.Sc. in Computer Science and Artificial Intelligence*

- Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Algorithm Design, Software Engineering, and Ethical AI (bias and privacy).

**AI.SPIRE: Applied AI and ML Systems** 2026

*LevelUp Economy & Istidama Consulting*

Python for AI – Applied NLP & Knowledge Systems

- 24-week intensive program covering applied NLP, machine learning, RAG systems, and Knowledge Graphs
- Built and deployed end-to-end AI systems integrating NLP pipelines, vector search, and APIs
- Developed professional skills in technical communication, teamwork, and AI system documentation

## KEY PROJECTS

### Meridian Retail — Agentic Customer Support System

**Tech:** Python, Weaviate, Neo4j, BM25, Groq LLMs, FastAPI, Docker, Redis, Next.js

- Designed and built a production-grade AI agent (not a scripted chatbot) using hybrid retrieval — dense vector search (Weaviate), BM25 keyword search, and a Neo4j knowledge graph — with Groq-hosted LLMs for intent classification, generation, and grounded fact-checking.
- **Achieved 96.7% routing accuracy and 95.9% groundedness** on a 60-example hand-labeled held-out set, improving routing from an 86.7% baseline through iterative, evidence-driven threshold tuning.
- **Raised escalation recall from 33% to 100%** on adversarial test cases with zero false-positive regressions across the full evaluation set, after diagnosing and fixing a critical escalation-detection gap.
- **Cut mean response time by 67% (3.32s → 1.08s)** by profiling and eliminating redundant embedding computation in the groundedness-check pipeline.
- **Built a 164-test automated suite** (unit + integration) covering agent routing logic, guardrails, retrieval, and escalation flows.
- Deployed the full stack to production across Weaviate Cloud, Neo4j AuraDB, Upstash Redis, Render, and Vercel, migrating from a fully local Docker environment to a live cloud deployment.
- Extended post-launch with Arabic language support (native query translation for cross-lingual retrieval), voice input, file attachments, persistent chat history, and a two-way human-handoff workflow via an internal admin dashboard.

### Smart Detection System for the Visually Impaired *(Graduation Project)*

**Tech:** Python, YOLOv8, Deep Learning, gTTS, Streamlit

- Engineered a real-time assistive navigation system that detects traffic signs, obstacles, pedestrians, and stairs using a YOLOv8 object-detection pipeline for live video processing.
- **Built the training dataset end to end (annotation, class balancing, augmentation) and trained a 16-class YOLOv8 detector on 250K images, reaching 80% mAP@0.5 and 70% F1.**
- Generated Arabic voice guidance via gTTS, conveying object type, direction, and distance to support safe, independent mobility for visually impaired users.
- Delivered an interactive Streamlit interface presenting live video with synchronized visual and audio feedback.

### Brain Tumor Classification ↗ Live Demo

**Tech:** Python, PyTorch, EfficientNet-B3, Grad-CAM, FastAPI, React, Docker

- **Fine-tuned an EfficientNet-B3 model to classify brain MRI scans into four classes (glioma, meningioma, pituitary, no tumor), reaching 95% test accuracy on 5,600 MRI scans.**
- Added Grad-CAM heatmaps to visualize the regions driving each prediction, improving interpretability for clinical review.
- Deployed as a full-stack live demo (FastAPI backend + React frontend, containerized with Docker) for real-time inference on uploaded MRI images.

### Voice & Facial Emotion Recognition System ↗ Live Demo

**Tech:** Python, PyTorch, ResNet-18, BiLSTM, Vision Transformer (ViT), Wav2Vec2

- Built a multimodal emotion-recognition system fusing facial-expression and voice-emotion analysis to classify five emotions (happy, sad, angry, fearful, neutral), with switchable custom-trained and pretrained-foundation inference paths.
- **Trained on 9,227 audio samples (RAVDESS, CREMA-D, TESS) and 31,338 face images (FER2013), reaching 76.86% accuracy with late fusion of the custom-trained models.**
- Engineered a rich acoustic feature set (MFCC, log-mel, ZCR, and prosodic features: F0, jitter, shimmer, HNR) and delivered a live Streamlit demo with real-time webcam + mic streaming.

### Multilingual Intent & NER Chatbot ↗ Live Demo

**Tech:** Python, DistilBERT, FAISS, Qwen2.5, Gradio

- Developed a multilingual customer-service chatbot supporting Arabic, English, French, and code-switching, with automatic language detection, intent classification, NER, FAISS retrieval, and LLM-generated replies.
- **Achieved 92.45% intent-classification accuracy, 99.62% language-detection accuracy, and 0.8455 NER F1 on a held-out test set, with a 1.78s median response latency.**
- Implemented cross-lingual retrieval (query in Arabic, retrieve from an English knowledge base) and deployed an interactive Gradio interface with token-by-token streaming.

### Neural Digest — Automated AI Content Bot ↗ Live Channel

**Tech:** Python, Groq (llama-3.3-70b), GitHub Actions, Telegram Bot API, arXiv/RSS, Pillow

- **Built a fully automated Telegram bot that publishes 8 daily educational AI posts (arXiv papers, news, and technical concepts) — running entirely on GitHub Actions with zero server infrastructure.**
- Engineered an end-to-end pipeline: source fetching (arXiv + RSS), Groq-powered Arabic explanation and summarization, JSON-based deduplication, and Pillow-generated image cards, with automatic fallback modes when no fresh content is available.

### Laptop Advisor — Recommendation Assistant ↗ Live Demo

**Tech:** Python, Gradio, Hugging Face Spaces

- **Built an AI recommendation assistant for the Jordanian laptop market that matches users to suitable options from a 150-laptop catalog based on budget, use case, and target specifications.**
- Deployed as a publicly accessible live demo on Hugging Face Spaces for real-time, on-demand recommendations.

### Heart Disease Prediction

**Tech:** Python, Scikit-learn, Random Forest, SVM, Data Analysis

- **Built and evaluated Random Forest and SVM classifiers to predict heart-disease risk from clinical indicators on a 70,000-patient dataset, reaching 88% accuracy, with data cleaning, feature analysis, and model evaluation.**

### Telecom Customer Churn Prediction

**Tech:** Python, Scikit-learn, Random Forest, Ensemble Methods, Data Analysis

- Built an F1-focused churn-prediction pipeline on a ~4,500-customer telecom dataset, using Random Forest hyperparameter tuning, nested cross-validation, and ensemble methods to produce reliable, generalizable estimates for business decision-making.

## TECHNICAL SKILLS

- **Languages & Databases:** Python, SQL, PostgreSQL, SQLite
- **ML & Frameworks:** Machine Learning, Deep Learning, Neural Networks, Regression, Decision Trees, Random Forests, Class-Imbalance Handling; TensorFlow, PyTorch, Scikit-learn, CNNs, EfficientNet, Vision Transformers (ViT)
- **Computer Vision & NLP:** YOLOv8, Object Detection, Image Classification, Grad-CAM, Data Augmentation; Transformers (BERT/RoBERTa/GPT/Qwen), Embeddings (word2vec/BERT/SBERT), spaCy, Hugging Face, NER, Multilingual & Code-Switching NLP, Wav2Vec2
- **Retrieval, Data & BI:** RAG Pipelines, Vector Databases (Weaviate, FAISS), Knowledge Graphs (Neo4j), Triple Stores (Fuseki), SPARQL; Statistics, Hypothesis Testing, EDA, Predictive Modeling, Advanced Excel, Tableau, Looker Studio, KPI Design, Matplotlib, Seaborn, Plotly

- **MLOps & Tools:** FastAPI, Docker, Docker Compose, Next.js, GitHub Actions, Hugging Face Spaces, Streamlit, Gradio, Git, JupyterLab, Unit Testing, Agentic AI-assisted IDEs

## CERTIFICATIONS & TRAINING

**Advanced Data Analytics — Tech for Jobs Program · Correlation One (USAID)** Mar 2025

- Graduated with Honors · 250 hours — comprehensive applied data analytics program covering Excel, analytical SQL, Python (NumPy/Pandas), statistics & EDA, and BI (Tableau, Looker Studio), with an end-to-end capstone.

**Python AI (Machine Learning) — The Hope International Company** Oct 2024

**ASP.NET (MVC) — The Hope International Company** Dec 2023

## ADDITIONAL

- **Soft Skills:** Problem Solving, Critical Thinking, Adaptability, Creativity, Communication & Presentation, Leadership
- **Languages:** Arabic (Native), English (Professional Working Proficiency)